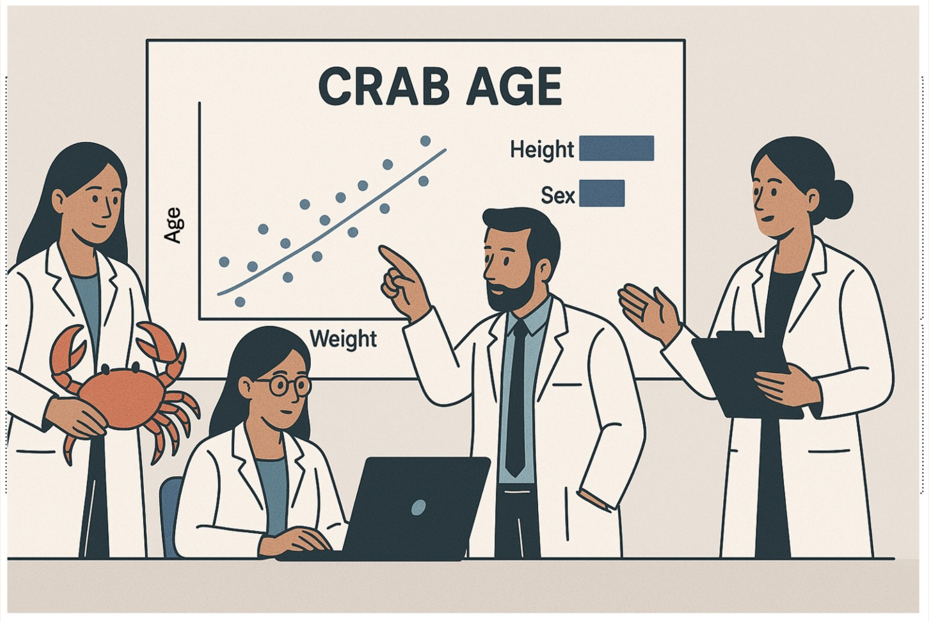
PREDICTING THE AGE OF CRABS



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GOALS

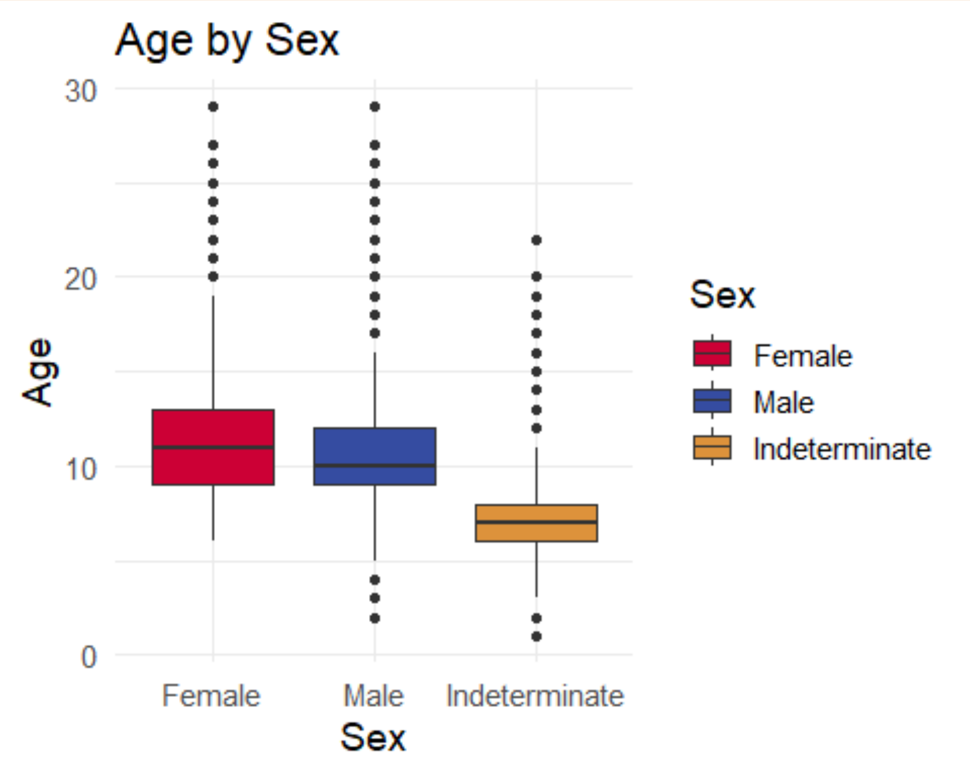
- Identify the key determinants of crab age.
- Explore key factors or relationships that contribute to predicting the age of crabs.

DATASET

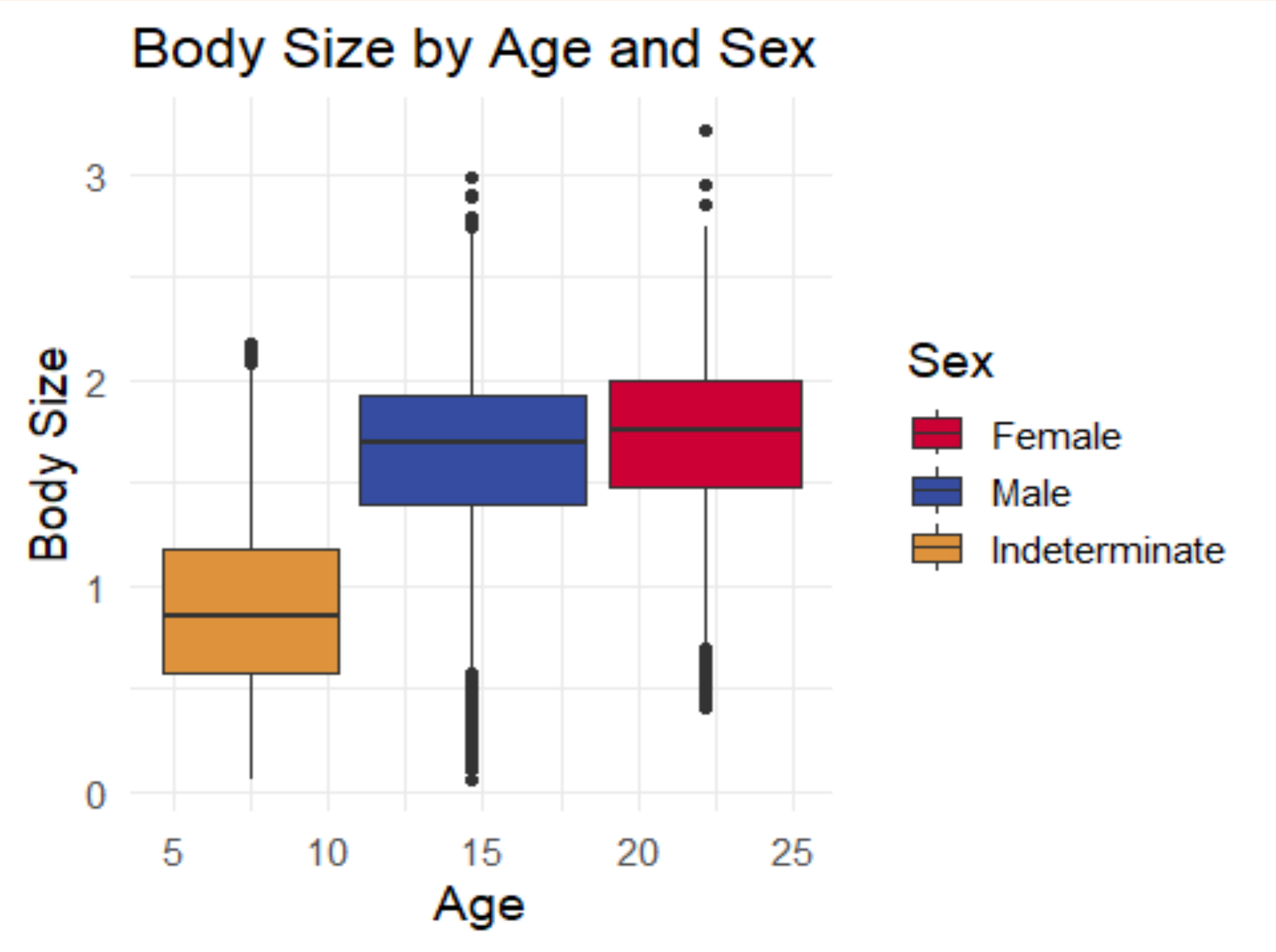
FEATURE DETAILS

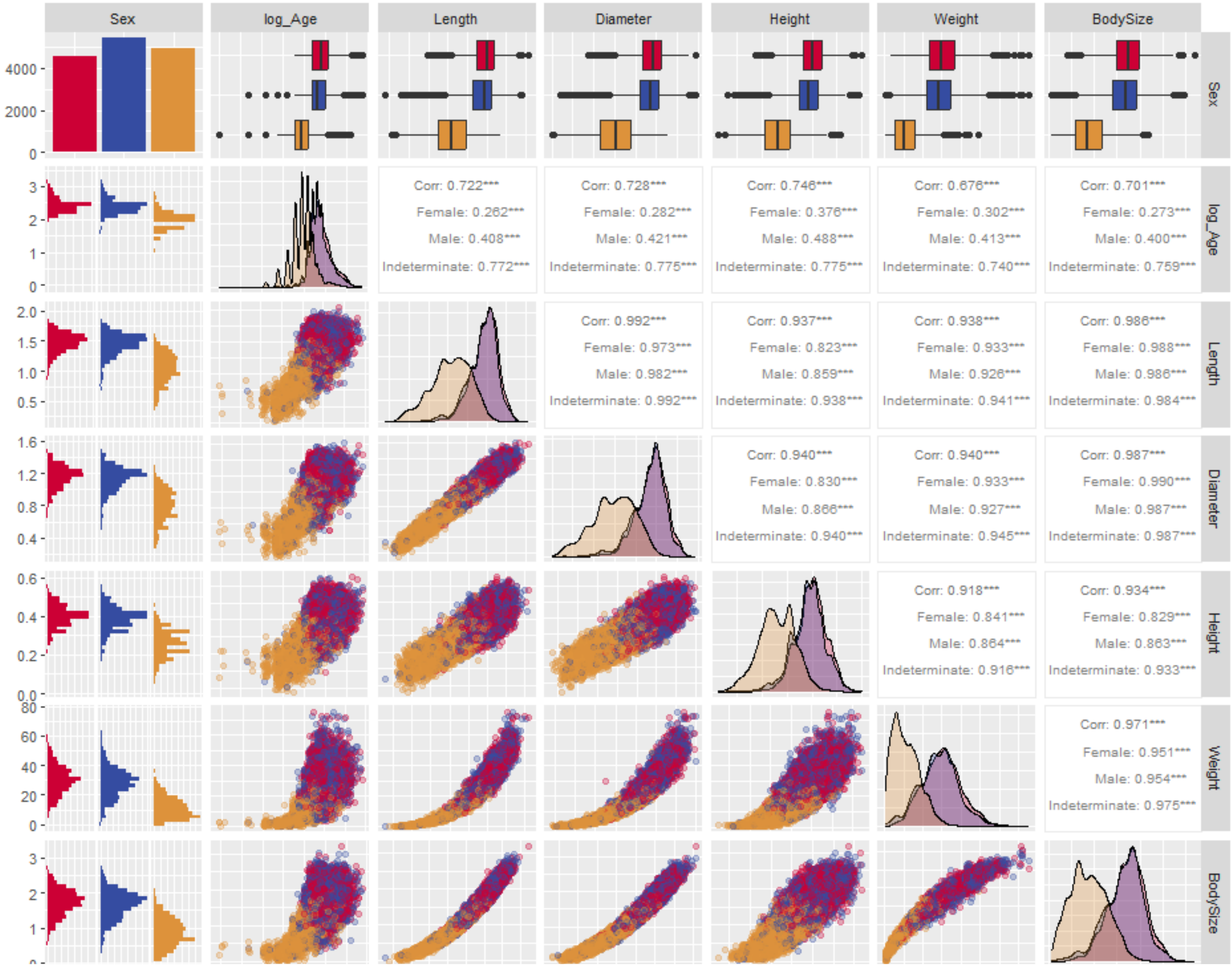
- Sex (M, F, or Indeterminate)
- Length
- Diameter
- Height
- Weight
- Shucked Weight
- Viscera Weight
- Shell Weight

Output Variable: Age (in months)

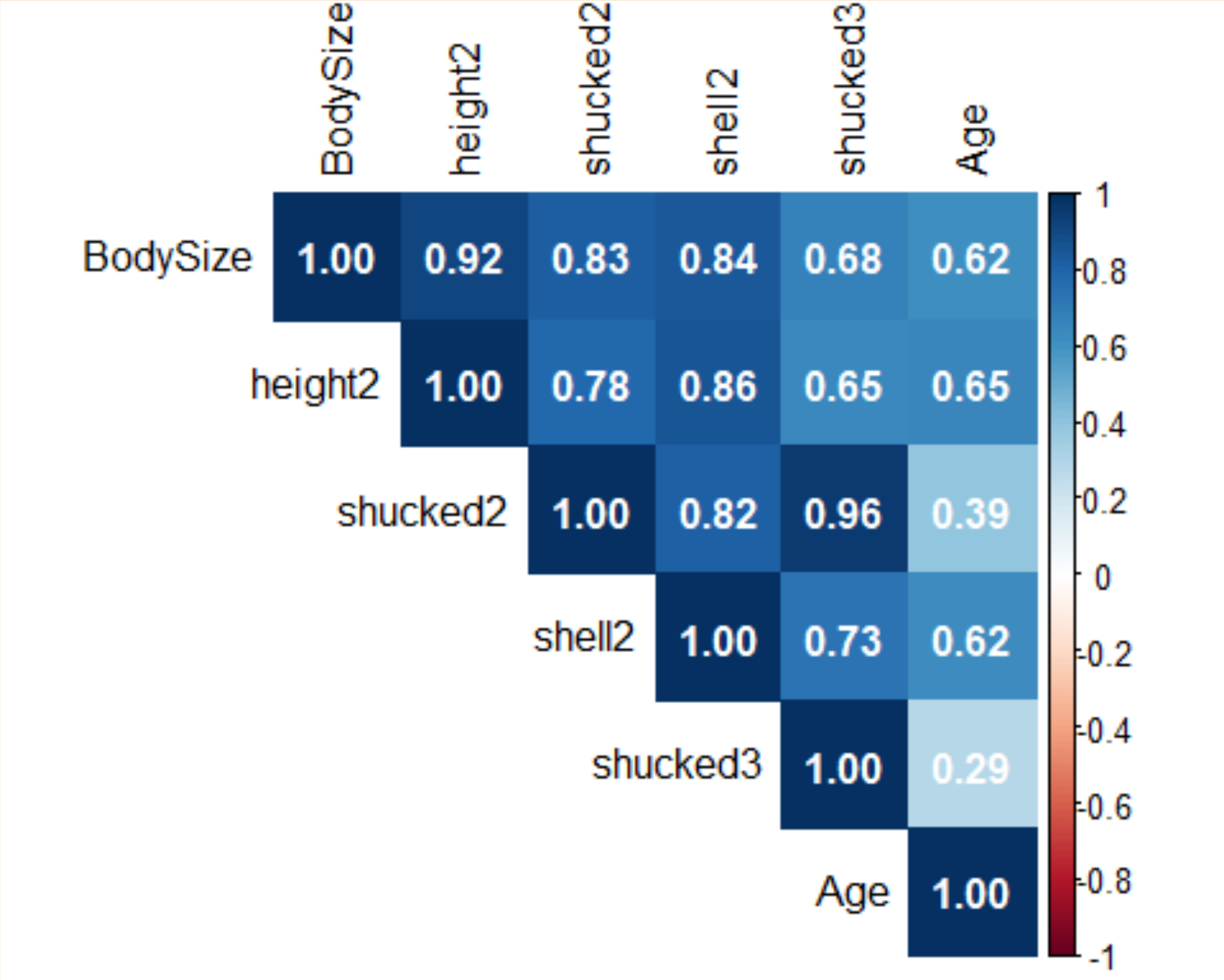


Does the relationship between size and age depend on sex?





Sex
log_Age
Length
Diameter
Height
Weight
BodySize
4000
2000
0
3
2
1
0
2.0
1.5
1.0
0.5
1.6
1.2
0.8
0.4
0.6
0.4
0.2
0.0
80
60
40
20
0
3
2
1
Corr: 0.722***
Female: 0.262***
Male: 0.408***
Indeterminate: 0.772***
Corr: 0.728***
Female: 0.282***
Male: 0.421***
Indeterminate: 0.775***
Corr: 0.746***
Female: 0.376***
Male: 0.488***
Indeterminate: 0.775***
Corr: 0.676***
Female: 0.302***
Male: 0.413***
Indeterminate: 0.740***
Corr: 0.701***
Female: 0.273***
Male: 0.400***
Indeterminate: 0.759***
Corr: 0.992***
Female: 0.973***
Male: 0.982***
Indeterminate: 0.992***
Corr: 0.937***
Female: 0.823***
Male: 0.859***
Indeterminate: 0.938***
Corr: 0.938***
Female: 0.933***
Male: 0.926***
Indeterminate: 0.941***
Corr: 0.986***
Female: 0.988***
Male: 0.986***
Indeterminate: 0.984***
Corr: 0.940***
Female: 0.830***
Male: 0.866***
Indeterminate: 0.940***
Corr: 0.940***
Female: 0.933***
Male: 0.927***
Indeterminate: 0.945***
Corr: 0.987***
Female: 0.990***
Male: 0.987***
Indeterminate: 0.987***
Corr: 0.918***
Female: 0.841***
Male: 0.864***
Indeterminate: 0.916***
Corr: 0.934***
Female: 0.829***
Male: 0.863***
Indeterminate: 0.933***
Corr: 0.971***
Female: 0.951***
Male: 0.954***
Indeterminate: 0.975***


BodySize
height2
shucked2
shell2
shucked3
Age
1.00
0.92
0.83
0.84
0.68
0.62
1.00
0.78
0.86
0.65
0.65
1.00
0.82
0.96
0.39
1.00
0.73
0.62
1.00
0.29
1.00
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1

Best MAE:

Sex

Shucked.Weight

Diameter

Height

Weight

Shell.Weight

Viscera.Weight

Height^2

Weight^2

Shucked.Weight^2

Viscera.Weight^2

Shell.Weight^2

Shell.Weight^3

Sex Shucked.Weight

Diameter × Height

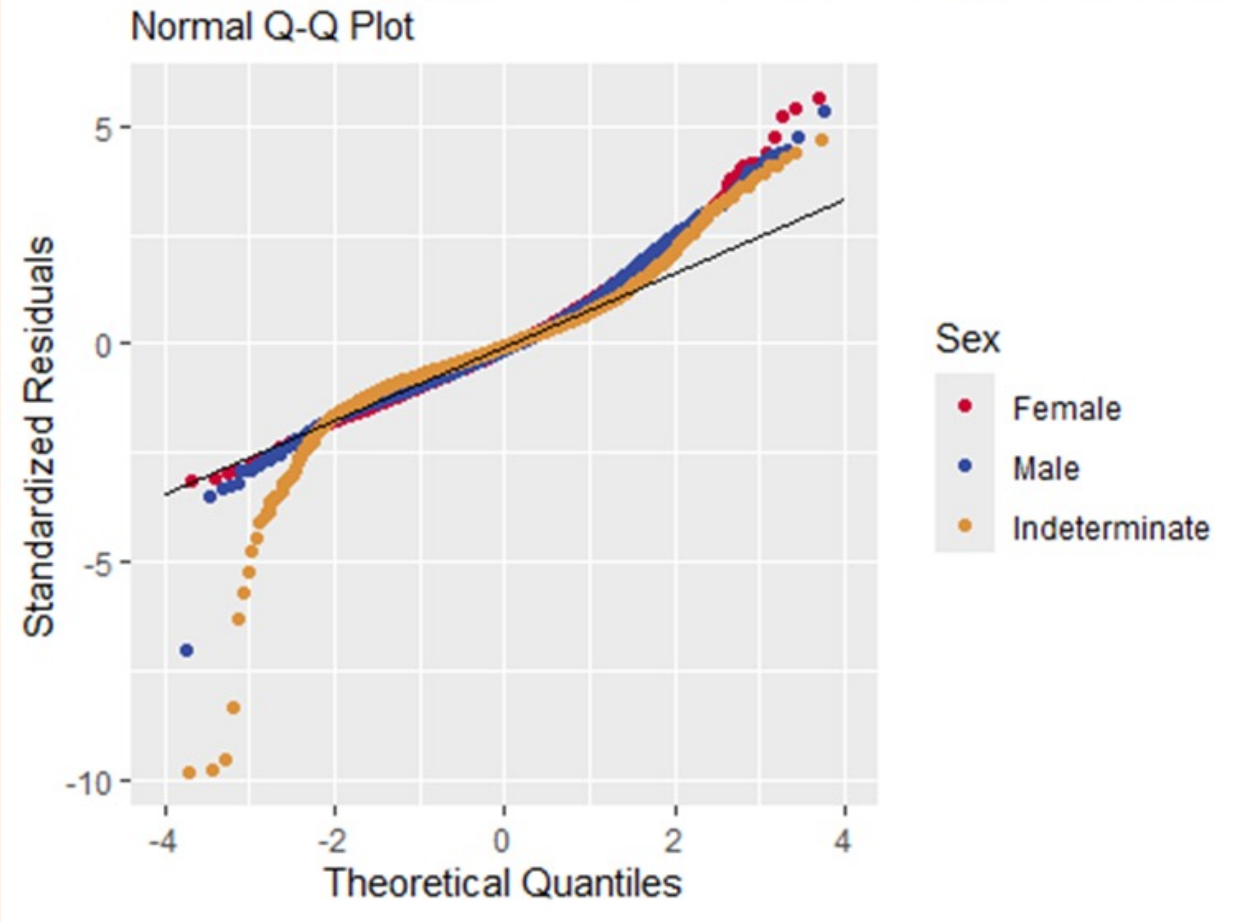
Shell.Weight × Weight

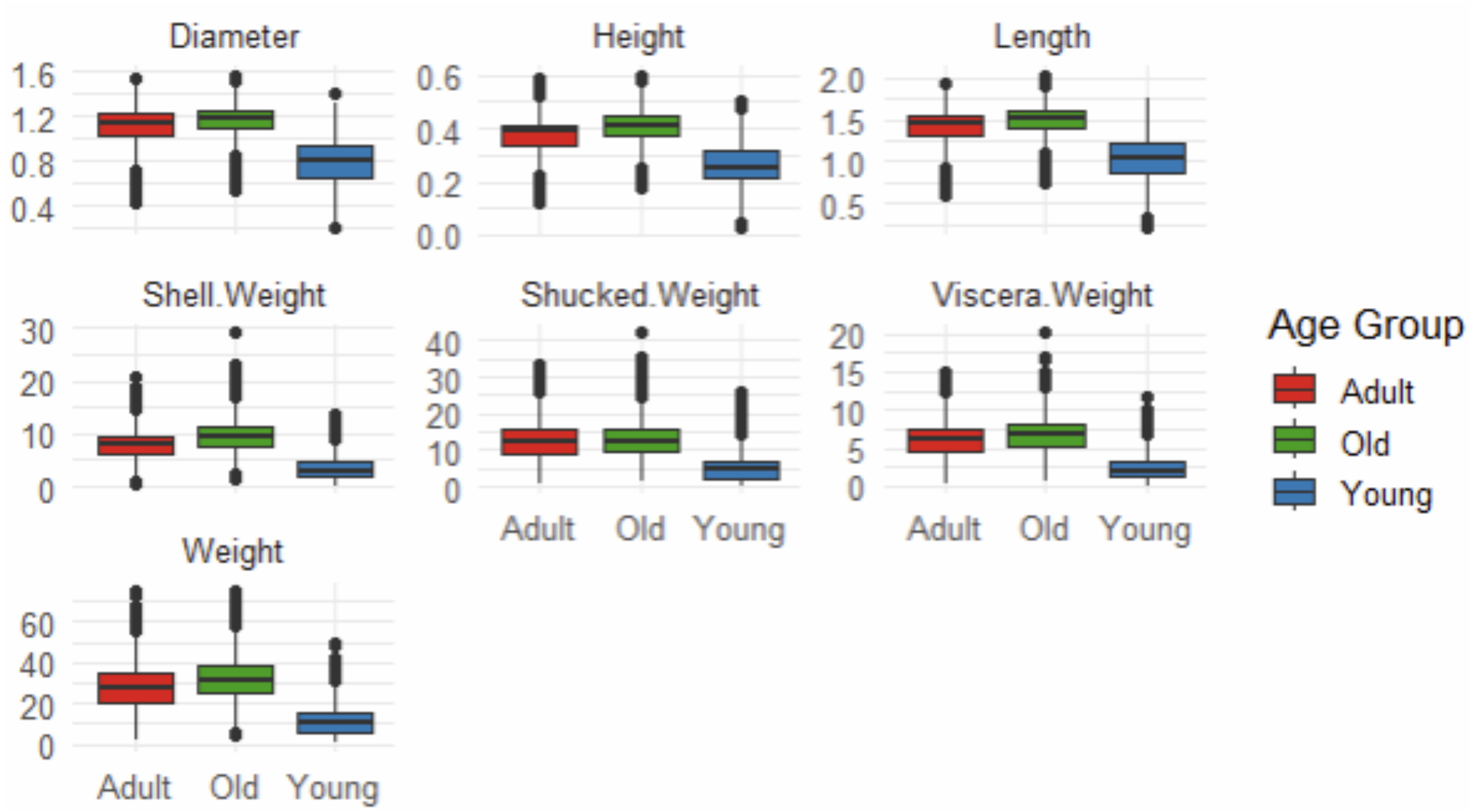
Shell.Weight × Shucked.Weight

Shell.Weight × Viscera.Weight

ShellDensity

R^2 = 70.2%
Adjusted R^2 = 70.16%
MAE = 1.14





Diameter
Height
Length
Shell.Weight
Shucked.Weight
Viscera.Weight
Weight
Adult
Old
Young
Age Group

Crab Analysis | **Overview**

Filters

Sex: All

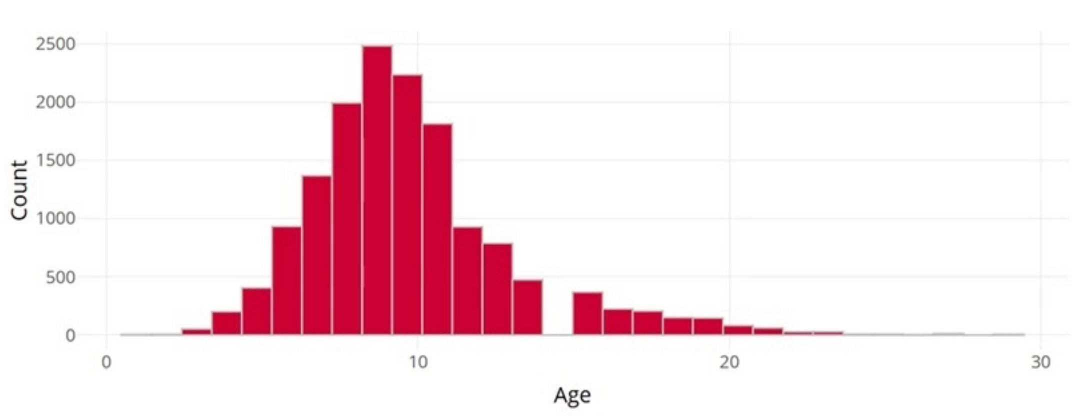
Age: 1 – 29

Weight: 0.23 – 75.32

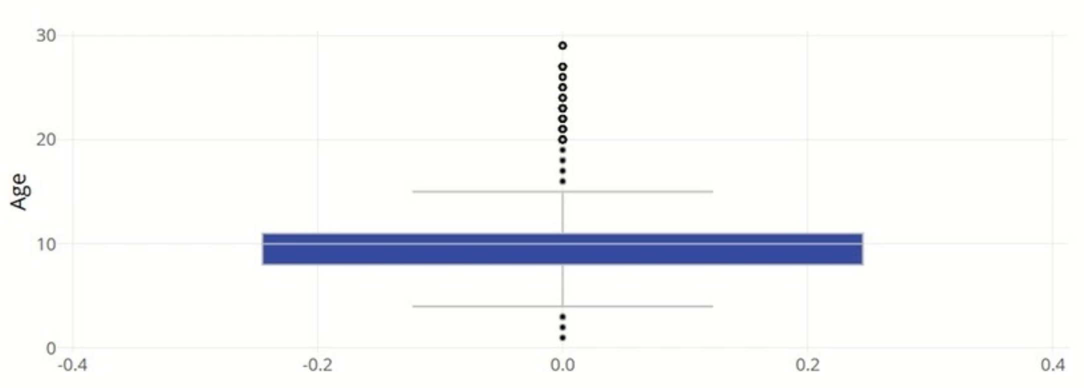
Reset Page

Average Age	Average Weight	Average Height
10	23.24	0.35

Age Histogram



Age Box Plot



Crab Data

Sex	Length	Diameter	Height	Weight	Shucked Weight	Viscera Weight	Shell Weight	Age
F	1.38	1.05	0.29	20.88	9.82	4.82	5.1	9
M	1.24	0.95	0.31	16.84	7.54	3.71	4.39	10
M	1.61	1.26	0.4	37.99	15.97	7.98	13.61	18
I	1.15	0.86	0.29	10.45	4.85	2.37	3.54	8
I	0.94	0.69	0.21	6.8	2.95	1.35	1.98	6
F	1.48	1.21	0.41	35.86	13.93	7.98	10.77	11
M	1.38	1.11	0.36	26.08	11.48	6.38	7.65	10

Model 1 Summary

MAE (original scale): 1.15

Statistic	Value
R	0.809
R-Squared	0.655
Adj. R-Squared	0.655
Pred R-Squared	0.653
MAE	**0.14**
RMSE	0.189

Model 2 Summary

MAE (original scale): 1.14

Statistic	Value
R	0.835
R-Squared	0.698
Adj. R-Squared	0.697
Pred R-Squared	0.696
MAE	**0.131**
RMSE	0.177

THANK YOU!

Chloe Barker, Drew Nunnally, Tracy Dower